tara.friedrich@gmail.com

Tara A Friedrich

tara-friedrich-bioinformatics

+1 (408) 242-0551 San Francisco, CA staracode

Skills

- ***Programming:*** Python, R, bash, MySQL
- ***Data Science:*** pandas, numpy, seaborn, matplotlib, ggplot2, scikit-learn, scikit-image
- ***Computational Resources:*** high performance cluster, AWS, git version control
- ***Statistics:*** Hypothesis testing, linear and generalized linear modeling, clustering, dimensionality reduction, classification, boosting, random forests, model selection
- ***Genomics:*** single-cell and bulk RNA-seq, ChIP-seq, microarrays, SNP analysis, ATAC-seq

Experience

Bioinformatics Programmer III at UCSF Liver Center July 2018–July 2019

- Used various regression and machine learning models to predict genes involved in liver development.
- Identified possible candidate variants predictive of NASH in patients using statistical tools and models.
- Independently organized a system to track and process various datasets from multiple research labs.
- Wrote code to process datasets and used version control to track changes so results were reproducible.

Health Data Science Fellow at Insight Jan 2018–May 2018

- Developed a Python machine learning model for *EpiBiome* to predict bacterial targets of phage.

Bioinformatician at the Gladstone Institutes Mar 2017–Nov 2017

- Successfully designed research plans with estimated costs and timelines to meet targets for 20+ clients.
- Interpreted and communicated results through presentations, figures, documentation, and in-person discussions to address client needs.
- Developed initial pipeline to identify if an experimental protocol (RASL-seq) passed quality control and worked with researchers to develop a better protocol.

Intern at DNAnexus Jun 2016–Sep 2016

- Tested methods to scale and parallelize 10x Genomics's *Long Ranger* alignment software across AWS instances to expand the customer's tool base.
- Reported to supervisors ways that *Long Ranger* could be distributed across AWS and communicated with third-party (10x Genomics) about how I could improve this use of their software.
- Stress-tested *Long Ranger* /*Supernova* alignment software using genomic datasets (different sizes and memory requirements) to fix bottlenecks and pushed changes to shared git codebase.

Graduate Student Researcher at the Pollard Lab, UCSF Sep 2011–Mar 2017

- Performed phylogenetic statistical analysis to discover fast-evolving regulatory sequences to calculate the likelihood that an increased number of DNA substitutions was greater than expected (link).
- Worked on novel statistical package (*motifDiverge* in R) to detect differences in transcription factor binding sequence motifs that considers the relationship between evolutionarily related sequences (link).
- Adapted package *CQN* in R to allow gene expression comparisons between multiple species to control for differences in GC content and gene length between species (link).

Education

- Ph.D. Bioinformatics University of California, San Francisco 2011–2017
- BSc. Biochemistry University of California, Los Angeles 2002–2006